PROJECT-2: CREDIT CARD TRANSACTIONS ANALYTICS

PROJECT SUMMARY REPORT

Report:

Credit Card Transactions Analytics Using SAS Studio

Item	Details
Prepared from	SAS code, Excel output workbook, source CSV dataset, and project statement PDF
Dataset period	2013-10-04 to 2015-05-26
Source rows reviewed	26,052
Output workbook reviewed	11 sheets; tabular outputs only; no embedded charts/images
Prepared on	2026-07-02

1. Project Title

Credit Card Transactions Analytics in India Using SAS Studio

2. Project Overview

This project analyzes credit card transactions made across Indian cities. The completed implementation uses SAS Studio to import the transaction dataset, prepare clean analysis fields, solve the nine tasks specified in the project statement, and export the final results into an Excel workbook with separate result sheets.

The submitted files show a descriptive analytics project focused on spend concentration, card-type behavior, expense category patterns, gender-based contribution, month-over-month growth, weekend transaction behavior, and transaction velocity by city.

3. Problem Statement

The project statement requires analytical queries over credit card transactions in India. The expected outputs include top spending cities, highest-spend months by card type, cumulative spending milestones, Gold-card city contribution, highest and lowest expense categories by city, female spend contribution, January 2014 month-over-month growth, weekend spend ratios, and the fastest city to reach its 500th transaction.

No.	Task from project statement
1	Write a query to print top 5 cities with highest spends and their percentage contribution of total credit card spends.
2	Write a query to print highest spend month and amount spent in that month for each card type.
3	Write a query to print the transaction details (all columns from the table) for each card type when it reaches a cumulative of 1000000 total spends (We should have 4 rows in the o/p one for each card type).
4	Write a query to find city which had lowest percentage spend for gold card type.
5	Write a query to print 3 columns: city, highest_expense_type , lowest_expense_type (example format : Delhi , bills, Fuel).
6	Write a query to find percentage contribution of spends by females for each expense type.
7	Which card and expense type combination saw highest month over month growth in Jan-2014.
8	During weekends which city has highest total spend to total no of transaction’s ratio?
9	Which city took least number of days to reach its 500th transaction after first transaction in that city?

4. Project Objectives

- Import and standardize the credit card transaction dataset in SAS.
- Validate data completeness, duplicates, date parsing, and transaction amount ranges.
- Apply grouped aggregation, conditional aggregation, date logic, cumulative calculations, and BY-group processing.
- Generate task-wise result tables that directly answer all nine project questions.
- Produce a professional Excel output workbook suitable for submission and review.

5. Dataset Description

Dataset attribute	Observed value
File reviewed	Credit card transactions - Project - 2.csv
Total records	26,052
Fields	index, City, Date, Card Type, Exp Type, Gender, Amount
Missing values	0
Duplicate rows	0
Date parse failures	0
Date range	2013-10-04 to 2015-05-26
Unique cities	986

Card types	Gold, Platinum, Signature, Silver
Expense types	Bills, Entertainment, Food, Fuel, Grocery, Travel
Gender values	F, M
Total spend	4,074,833,373
Average transaction amount	156,411.54
Minimum / maximum amount	1,005 / 998,077

6. Methodology and Workflow

- The CSV file is imported into SAS using PROC IMPORT with maximum row scanning for reliable type inference.
- A clean working table is created with standardized variable names, parsed SAS dates, transaction month, weekday number, and normalized amount fields.
- PROC SQL is used for most grouped summaries, percentages, and conditional aggregations.
- PROC SORT and DATA step BY-group logic are used where ordered calculations are required, including cumulative card spend and city transaction sequencing.
- ODS EXCEL exports profile, quality checks, and task outputs into a multi-sheet workbook.

7. SAS Procedures, Techniques, and Analytical Methods Used

Technique	How it was used
PROC IMPORT	Loaded the CSV transaction data into SAS.
DATA step	Renamed variables, parsed dates, prepared month and weekday fields, and calculated ordered values.
PROC SQL	Used for grouped summaries and result creation; 8 PROC SQL blocks were identified in the code.
PROC SORT	Used for BY-group processing and ranking logic; 9 sort steps were identified.
CASE WHEN	Calculated female spend contribution inside expense-category groups.
Cumulative total	Calculated running spend by card type to find the first transaction crossing 1,000,000.
Month-over-month logic	Compared January 2014 combination spend against the previous month.
ODS EXCEL	Created the final task-wise Excel workbook.

8. Data Cleaning and Preprocessing Steps

- Source fields were standardized into readable analysis variables such as city, card_type, expense_type, gender, txn_date, and amount.
- The original transaction date field was converted into a SAS date and formatted for analysis.
- A month-start variable was created for monthly aggregation.

- A weekday variable was created so Saturday and Sunday transactions could be filtered for the weekend analysis.
- The submitted output confirms zero missing values and zero duplicate transaction identifiers in the final quality check sheet.

9. Statistical Analysis and Data Visualization Performed

The completed analysis is tabular and statistical in nature. The Excel output workbook contains 11 sheets: dataset profile, quality checks, and nine task-specific output sheets. No embedded charts, graph objects, or image-based visualizations were found in the submitted Excel workbook.

The statistical analysis includes total spend, percentage contribution, category-level contribution, monthly spend ranking, cumulative spend milestone tracking, spend-per-transaction ratio, and transaction-count velocity.

10. Key Findings and Insights

Area	Finding
Top city spend	Greater Mumbai, India recorded the highest total spend at 576,751,476, contributing 14.1540% of total spend.
Spend concentration	The top four cities were Greater Mumbai, Bengaluru, Ahmedabad, and Delhi; each contributed more than 13% of total spend.
Highest month by card	Silver reached the largest card-month spend among task outputs, with 59,723,549 in March 2015.
Gold-card low contribution	Dhamtari, India had the lowest Gold-card spend contribution, with spend of 1,416.
Female contribution	Bills had the highest female contribution percentage among expense categories at 63.9459%.
MoM growth	The highest January 2014 month-over-month growth was Platinum + Grocery, increasing by 4,498,781.
Weekend ratio	Sonepur, India had the highest weekend spend-per-transaction ratio at 299,905; this was based on one weekend transaction.
Transaction velocity	Bengaluru, India reached its 500th transaction fastest, taking 81 days.

11. Results and Outcomes

Task	City	Spend	Contribution
Task 1	Greater Mumbai, India	576,751,476	14.1540%
Task 1	Bengaluru, India	572,326,739	14.0454%
Task 1	Ahmedabad, India	567,794,310	13.9342%
Task 1	Delhi, India	556,929,212	13.6675%

Task 1	Kolkata, India	115,466,943	2.8337%

Highest spend month by card type:

Card type	Highest month	Spend
Gold	Jan-2015	55,455,064
Platinum	Aug-2014	57,936,507
Signature	Dec-2013	58,799,522
Silver	Mar-2015	59,723,549

Other task outcomes:

Task	Output
Task 3	First cumulative 1,000,000 crossing was identified for each card type; four output rows were produced.
Task 4	Lowest Gold contribution city: Dhamtari, India.
Task 5	Highest and lowest expense type was produced for 986 cities.
Task 6	Female spend percentage was produced for all six expense types.
Task 7	Highest Jan-2014 growth combination: Platinum + Grocery.
Task 8	Best weekend spend ratio city: Sonepur, India.
Task 9	Fastest city to 500 transactions: Bengaluru, India.

12. Challenges Faced and How They Were Addressed

- Path handling in SAS Studio required Linux-style paths beginning with /home/u64542512 rather than local Windows paths.
- The submitted SAS code still contains a backslash in the Excel output path macro. The workbook was produced, but the downloaded file name became outputscredit_card_project2_outputs.xlsx. The cleaner SAS Studio path is &output_folder./credit_card_project2_outputs.xlsx.
- The weekend ratio result is correct for the stated formula, but the leading city has only one weekend transaction. A business version of the analysis could add a minimum transaction-count threshold.
- No separate SAS log file, PDF report, permanent SAS dataset file, or graph image was uploaded for this review. The summary therefore focuses on the SAS code, source CSV, problem statement, and Excel output workbook actually provided.

13. Business Impact or Practical Applications

- City-level spend concentration helps prioritize market, partnership, and customer engagement strategies.
- Card-type monthly peaks can support campaign timing and card portfolio planning.

- Expense-category contribution patterns help identify where customer segments are most active.
- Weekend spend ratios can identify high-value outlier transactions, but should be paired with transaction-count thresholds for operational decisions.
- Transaction velocity by city can indicate adoption speed and intensity of credit card usage.

14. Conclusion

The completed SAS project successfully addresses all nine analytical tasks from the project statement. The implementation demonstrates practical use of SAS data preparation, SQL-style aggregation, date handling, conditional logic, cumulative calculations, BY-group processing, and Excel reporting. The output workbook confirms that the dataset was imported, validated, and summarized into clear task-wise result tables.

Overall, the project is suitable for internship submission as a descriptive analytics case study, with one recommended cleanup: regenerate the Excel output using a forward slash in the SAS Studio output path so the downloaded workbook has a clean file name.

15. Future Scope and Recommendations

- Add visual dashboards or charts for city spend, card-type monthly trends, and expense-category distribution.
- Export a final PDF report directly from SAS or prepare a separate formal report for submission.
- Save important WORK datasets into a permanent SAS library when the evaluator needs SAS dataset files.
- Add CSV exports for task outputs if the company requires machine-readable result files.
- Add threshold-based variants for ratio metrics to reduce one-transaction outlier effects.
- Include the SAS log in the final submission package to prove successful execution without errors.

Reviewed Submission Files

File	Review observation
Credit Card Transactions Code.sas	440-line SAS implementation using PROC IMPORT, PROC SQL, PROC SORT, DATA steps, PROC PRINT, and ODS EXCEL.
outputscredit_card_project2_outputs.xlsx	Excel workbook with 11 output sheets: profile, quality, and nine task sheets. No charts or images detected.
Credit card transactions - Project - 2.csv	Source transaction dataset with 26,052 records and no missing or duplicate rows detected.
Problem Statement - Project 2.pdf	Two-page project statement describing dataset fields and nine analytical tasks.